

TASK 3: High – Student Management System

C Programming – Code

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#include <stdio.h>
#include <string.h>

#define MAX 100

struct Student {
    int rollNo;
    char name[50];
    float marks;
};

void addStudent(struct Student s[], int *count) {
    printf("Enter Roll Number: ");
    scanf("%d", &s[*count].rollNo);

    printf("Enter Name: ");
    scanf(" %49[^\n]", s[*count].name);

    printf("Enter Marks: ");
    scanf("%f", &s[*count].marks);

    (*count)++;
    printf("Student added successfully.\n");
}

void displayStudents(struct Student s[], int count) {
    int i;

    if (count == 0) {
        printf("No student records found.\n");
        return;
    }

    printf("\n--- Student Records ---\n");
    for (i = 0; i < count; i++) {
        printf("Roll No: %d\n", s[i].rollNo);
        printf("Name   : %s\n", s[i].name);
        printf("Marks  : %.2f\n\n", s[i].marks);
    }
}

void searchStudent(struct Student s[], int count) {
    int roll, i, found = 0;

    printf("Enter Roll Number to search: ");
    scanf("%d", &roll);

    for (i = 0; i < count; i++) {
        if (s[i].rollNo == roll) {
            printf("Roll No: %d\nName: %s\nMarks: %.2f\n",
                s[i].rollNo, s[i].name, s[i].marks);
            found = 1;
            break;
        }
    }

    if (!found)
        printf("Student not found.\n");
}

void updateStudent(struct Student s[], int count) {
    int roll, i, found = 0;
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printf("Enter Roll Number to update: ");
scanf("%d", &roll);

for (i = 0; i < count; i++) {
    if (s[i].rollNo == roll) {
        printf("Enter new Name: ");
        scanf(" %49[^\n]", s[i].name);

        printf("Enter new Marks: ");
        scanf("%f", &s[i].marks);

        printf("Record updated successfully.\n");
        found = 1;
        break;
    }
}

if (!found)
    printf("Student not found.\n");
}

void deleteStudent(struct Student s[], int *count) {
    int roll, i, j, found = 0;

    printf("Enter Roll Number to delete: ");
    scanf("%d", &roll);

    for (i = 0; i < *count; i++) {
        if (s[i].rollNo == roll) {
            for (j = i; j < *count - 1; j++)
                s[j] = s[j + 1];

            (*count)--;
            printf("Record deleted successfully.\n");
            found = 1;
            break;
        }
    }

    if (!found)
        printf("Student not found.\n");
}

int main() {
    struct Student students[MAX];
    int count = 0, choice;

    do {
        printf("\n--- Student Management System ---\n");
        printf("1. Add Student\n");
        printf("2. Display Students\n");
        printf("3. Search Student\n");
        printf("4. Update Student\n");
        printf("5. Delete Student\n");
        printf("6. Exit\n");
        printf("Enter your choice: ");
        scanf("%d", &choice);

        switch (choice) {
            case 1: addStudent(students, &count); break;
            case 2: displayStudents(students, count); break;
            case 3: searchStudent(students, count); break;
            case 4: updateStudent(students, count); break;
            case 5: deleteStudent(students, &count); break;
            case 6: printf("Exiting...\n"); break;
            default: printf("Invalid choice.\n");
        }
    } while (choice != 6);
}

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    return 0;  
}
```